

A compact-multiplier obstruction for Tandori function spaces

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Source Problem

This packet concerns Problem 7.5 of Kiwerski–Tomaszewski [1], page 51 of the source PDF:

of symbols that generate strictly singular and nuclear multipliers.

7.6. Compact multipliers between Tandori function spaces. For a given Köthe function space X , by the **abstract Tandori function space** $\tilde{X}$ we mean a vector space of all real-valued measurable functions, say f , with $\tilde{f} \in X$, where $\tilde{f}$ is the **least decreasing majorant** of a given function f , that is, $\tilde{f}(x) := \operatorname{ess\,sup}_{t \geq x} |f(t)|$. We equip $\tilde{X}$ with the norm $\|f\|_{\tilde{X}} := \|\tilde{f}\|_X$. It is straightforward to see that the ideal $(\tilde{X})_o$ is trivial (see, for example, [LM15, Theorem 1(e)]; in fact, due to [KKM21, Proposition 4.8], the space $\tilde{X}$ even contains a lattice isometric copy of ℓ_∞). Thus, in light of Theorem 4.1 and Theorem 4.16, we want to pose the following

Problem 7.5. *What can we say about (weakly) compact multipliers between $\tilde{X}$ and $\tilde{Y}$?*

The problem asks what can be said about weakly compact and compact multipliers between abstract Tandori function spaces.

Proof Intuition

The Tandori norm sees the right endpoint of a set rather than its exact location. If a measurable set $A \subset (0, b)$ has essential supremum b , then the least decreasing majorant of χ_A is $\chi_{(0, b)}$. Thus one can split any positive-measure level set of a nonzero symbol into infinitely many disjoint pieces with the same essential supremum. Their characteristic functions form a bounded sequence in $\tilde{X}$, while the images under multiplication by the symbol remain uniformly separated in $\tilde{Y}$. This rules out compactness.

Theorem

Let $I = (0, 1)$ or $I = (0, \infty)$, equipped with Lebesgue measure. Let X and Y be Köthe function spaces on I , in the convention of [1], and define the corresponding Tandori function spaces by

$$\tilde{X} = \{f : \tilde{f} \in X\}, \quad \|f\|_{\tilde{X}} = \|\tilde{f}\|_X,$$

where

$$\tilde{f}(x) = \operatorname{ess\,sup}_{t \geq x} |f(t)|.$$

If $M_\lambda : \tilde{X} \rightarrow \tilde{Y}$ is a compact multiplication operator, then $\lambda = 0$ almost everywhere.

Proof

Assume, toward a contradiction, that $\lambda \neq 0$ on a set of positive measure. Then for some $\varepsilon > 0$ the set

$$E = \{t \in I : |\lambda(t)| > \varepsilon\}$$

has positive measure. Since I is σ -finite, there is a finite number N such that $E \cap (0, N)$ has positive measure; in the case $I = (0, 1)$ take $N < 1$ after replacing E by a positive-measure intersection with a smaller interval if necessary. Put

$$B = E \cap (0, N), \quad b = \text{ess sup } B.$$

Then $0 < b < \infty$, and $B \cap (b - \delta, b)$ has positive measure for every $\delta > 0$.

Choose infinitely many pairwise disjoint measurable subsets $B_1, B_2, \dots \subset B$, each with essential supremum b . For completeness, here is one construction. Pick infinitely many disjoint positive-measure pieces in annuli approaching b , and partition the corresponding annulus indices into countably many infinite sets; the union of the pieces in the n -th infinite block is B_n . Non-atomicity lets us split the positive-measure annuli as needed.

For each n , since $\text{ess sup } B_n = b$, the least decreasing majorant of χ_{B_n} is exactly $\chi_{(0,b)}$. Set

$$u_n = \frac{\chi_{B_n}}{\|\chi_{(0,b)}\|_X}.$$

The source convention for Köthe spaces ensures $\chi_{(0,b)} \in X$, so $\|u_n\|_{\tilde{X}} = 1$.

If $n \neq m$, then $B_n \cup B_m \subset E$ is still a set with essential supremum b . Therefore

$$\lambda(\widetilde{u_n - u_m})(x) \geq \frac{\varepsilon}{\|\chi_{(0,b)}\|_X} \chi_{(0,b)}(x) \quad \text{for a.e. } x \in I.$$

By the ideal property of Y ,

$$\|M_\lambda u_n - M_\lambda u_m\|_{\tilde{Y}} \geq \varepsilon \frac{\|\chi_{(0,b)}\|_Y}{\|\chi_{(0,b)}\|_X} > 0.$$

Thus the bounded sequence (u_n) in $\tilde{X}$ has images under M_λ with no Cauchy subsequence in $\tilde{Y}$. This contradicts compactness of M_λ . Hence $\lambda = 0$ almost everywhere.

Verification And Scope

The proof is purely analytic and uses only the source paper's definition of Köthe function spaces and Tandori function spaces, plus the non-atomicity of the underlying interval. It answers the compact-multiplier part of Problem 7.5 in the standard function-space setting $I = (0, 1)$ or $I = (0, \infty)$. It does not settle the weakly compact multiplier part of the problem.

Novelty Check

Local run indexes contained no packet or attempt for arXiv:2212.06723 or for the Tandori compact-multiplier problem. Bounded web searches on 2026-06-14 for exact and close phrases such as "compact multipliers" "Tandori" function spaces, "Tandori function space" "compact multipliers", and "weakly compact multipliers" "Tandori" found the source paper but no later solution of Problem 7.5.

References

- [1] T. Kiwerski and J. Tomaszewski, *Essential norms of pointwise multipliers in the non-algebraic setting*, arXiv:2212.06723.